Rooftop Rain: turning school roofs into water banks

- Maya Okafor, Delft Water Lab

The problem

- Amsterdam schools flood 3-4 times a year during cloudbursts.
- Playgrounds are paved, so water has nowhere to go.

Our idea

- A shallow 8 cm storage layer under a green roof.
- Valves release water slowly after the storm.

Pilot results

- Four schools, 18 months.
- Peak runoff reduced by 62%.
- Roof temperature in summer down 9 degrees.

Cost

- EUR 115 per square metre installed.
- Pays back in roughly 12 years through lower drainage fees.

What's next

- Scaling to 40 schools in 2027 with the municipality.
- Open dataset released in March.